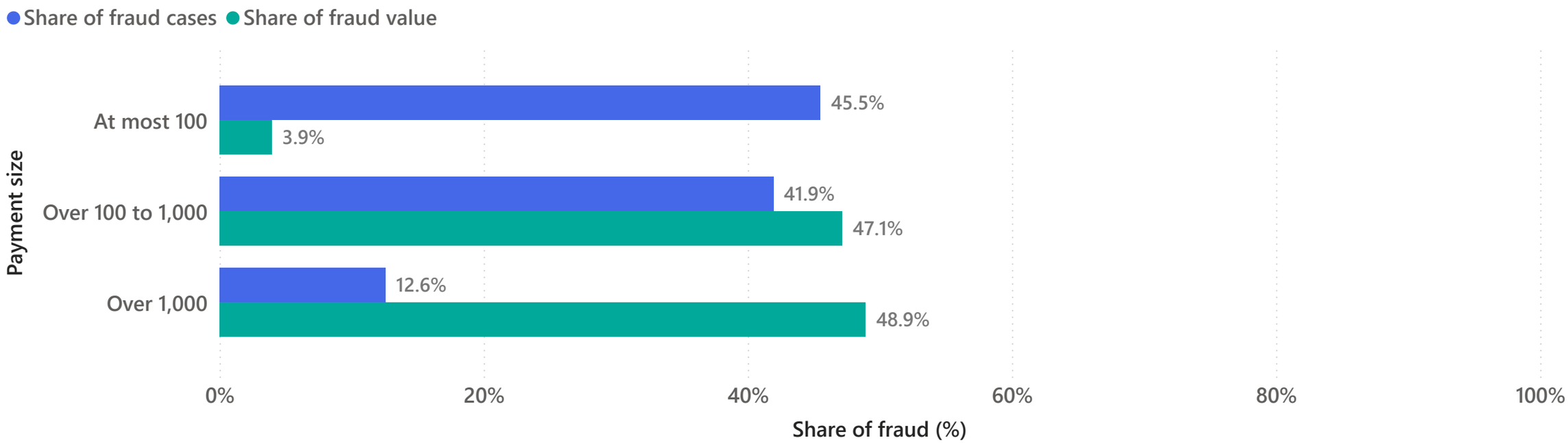
Fraud exposure and payment size

The complete 5-million-payment file; is_fraud is the recorded outcome.

Payments	Fraud cases	Fraud-labelled amount	Fraud rate
5.00M	179,553	64.37M	3.59%

Fraud cases and value by payment size



Amount in fraud-labelled payments is transaction value. Losses and recoveries are not supplied.

Payments above 1,000 account for 12.6% of fraud cases and 48.9% of fraud-labelled value. Payments of 100 or less account for 45.5% of cases but only 3.9% of value. Case volume and the amount under review therefore lead to different priorities.

Small payments drive case count; large payments drive value

Amount bands are lower-exclusive and upper-inclusive. Example: over 10 to 100 includes 100.

Fraud case mix and value mix

Amount band	All payments	Fraud cases	Fraud rate	Fraud value	Case share	Value share
0 to 10	440,823	15,701	3.56%	74,609.35	8.7%	0.1%
10 to 100	1,828,503	65,986	3.61%	2,463,740.38	36.8%	3.8%
100 to 500	1,472,134	52,836	3.59%	14,244,983.34	29.4%	22.1%
500 to 1,000	632,746	22,456	3.55%	16,103,014.09	12.5%	25.0%
1,000 to 2,500	624,125	22,517	3.61%	31,337,756.94	12.5%	48.7%
2,500 to 5,000	1,669	57	3.42%	150,709.70	0.0%	0.2%

Case share = band fraud cases / all fraud cases. Value share = band fraud amount / all fraud amount.

The 1,000-2,500 band alone contains 31.34 million of fraud-labelled value. Size does not clearly distinguish fraud from other payments: the median amount is 137.88 for fraud and 138.71 for legitimate labels. Amount is useful for sizing review exposure, rather than assigning a fraud verdict.

No sharp monthly or hourly fraud pattern appears

2023 full months. January 2024 contains one recorded day and is excluded from monthly comparisons.

2023 month	Payments	Fraud cases	Cases / day	Fraud rate
Jan 2023	418,302	14,770	476.5	3.53%
Feb 2023	382,551	13,777	492.0	3.60%
Mar 2023	425,454	15,247	491.8	3.58%
Apr 2023	411,054	14,761	492.0	3.59%
May 2023	423,451	15,133	488.2	3.57%
Jun 2023	410,890	14,759	492.0	3.59%
Jul 2023	424,493	15,395	496.6	3.63%
Aug 2023	425,546	15,208	490.6	3.57%
Sep 2023	411,277	14,855	495.2	3.61%
Oct 2023	424,906	15,331	494.5	3.61%
Nov 2023	411,100	14,730	491.0	3.58%
Dec 2023	424,559	15,376	496.0	3.62%

Observed rate ranges

Period	Low	High
Hours	3.51%	3.66%
Weekdays	3.56%	3.62%
2023 months	3.53%	3.63%

Hours use the recorded timestamp. A timezone is not provided.

The 1 January 2024 slice contains 6,417 payments and 211 fraud labels; it is not a second full year.

Fraud rates stay close across months and recorded hours. Time alone gives little reason to focus reviews on one part of the day. February has fewer cases partly because it has fewer days: the daily average ranges from 476.5 in January to 496.6 in July.

Rapid repeat payments explain very few recorded fraud cases

Gaps are measured after sorting each sender by timestamp; first observations stay separate.

Payments within one hour	Fraud cases within one hour	Share of all fraud cases	Median observed gap (days)
3,162	110	0.06%	36.9

Gap since previous sender payment	Payments	Fraud cases	Fraud rate	Fraud value
First observed payment	896,513	29,866	3.33%	10,665,437.70
Up to 60 seconds	49	2	4.08%	226.13
1 to 5 minutes	209	11	5.26%	2,518.68
5 to 60 minutes	2,904	97	3.34%	28,563.52
1 to 24 hours	72,249	2,778	3.85%	1,036,227.74
1 to 7 days	425,236	15,835	3.72%	5,646,928.22
Over 7 days	3,602,840	130,964	3.64%	46,994,911.81

These are gaps between payments visible in this file, not a complete account or card history.

Only 110 of 179,553 fraud cases follow the same sender's previous observed payment within one hour. A below-100 payment followed by one above 1,000 within 24 hours occurs 4,126 times, with 168 fraud labels (4.07%). That sequence merits checking, but it does not establish deliberate test charges.

Recorded-city changes have a higher observed fraud rate

Compare consecutive payments by the same sender. A first observation has no prior city to compare.

City changed	Same city	Difference (percentage points)	Changed / same-city rate
3.76%	2.89%	0.87	1.30x

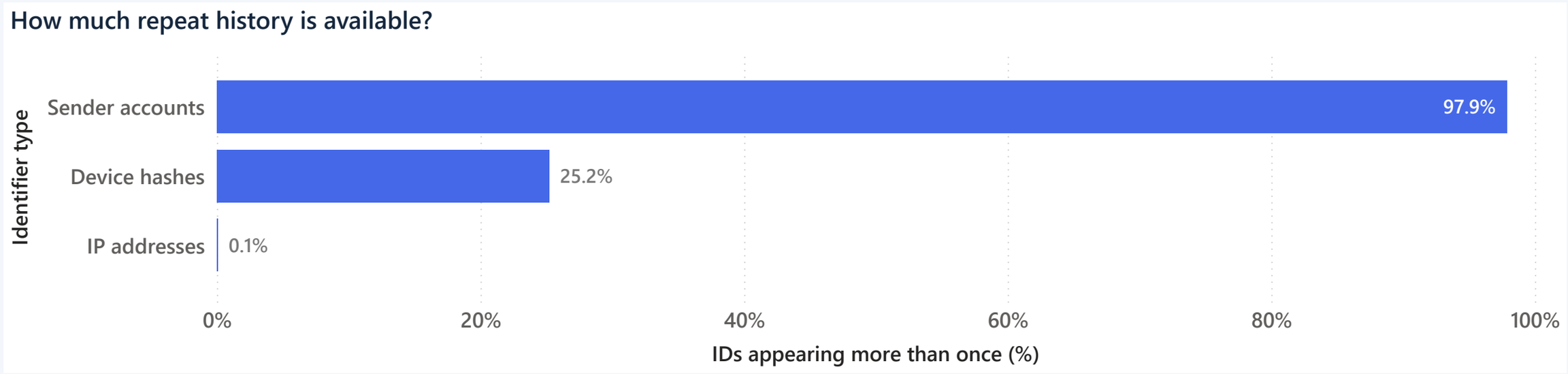
Sender history	Payments	Fraud cases	Fraud rate	Fraud value
Same location	514,269	14,840	2.89%	5,437,159.67
Location changed	3,589,218	134,847	3.76%	48,272,216.43
First observed payment	896,513	29,866	3.33%	10,665,437.70

A recorded city is not a verified home address or an IP-based location. Different transaction locations can also be legitimate.

The rate is 3.76% after a recorded-city change, versus 2.89% when the city stays the same: a difference of 0.87 percentage points, or 1.30 times the rate. This is a clearer comparison than ranking the eight cities, whose overall rates are almost identical. It remains an association in the supplied records.

Accounts repeat; most IP addresses do not

Reuse means more than one appearance anywhere in the complete file, not repeated fraud.



Identifier	Distinct IDs	Reused IDs	IDs with repeat fraud	Most frauds / ID
Sender accounts	896,513	877,303	16,337	7
Receiver accounts	896,639	877,383	15,604	5
Device hashes	3,835,723	967,776	1,757	3
IP addresses	4,997,068	2,932	6	2

Sender IDs provide substantially more repeat history than device hashes or IPs. Only six IP addresses have more than one fraud label. Fraud is also widely spread across accounts: the top ten senders by fraud count contain just 45 cases (0.025% of all fraud), and the top ten receivers contain 41.

Most supplied categories show little separation

Full-file observed fraud-rate ranges. Score ranges refer to the bands defined in the analysis.

Field compared	Lowest rate	Highest rate	Range (percentage points)
Payment channel	3.58%	3.60%	0.021
Transaction type	3.56%	3.63%	0.061
Merchant category	3.57%	3.61%	0.044
Recorded city	3.58%	3.60%	0.026
Device category	3.57%	3.62%	0.047
Velocity score bands	3.57%	3.61%	0.038
Spending score bands	3.57%	3.71%	0.136
Geographic score bands	3.57%	3.62%	0.051

Ranges describe the chosen groups; they are not a model test, a causal result, or a validated blocking threshold.

Channel, city and merchant rates are very close. The supplied velocity, spending-deviation and geographic scores also show little separation across these bands. None of these fields provides a clear standalone reason to prioritise one group. Review combinations of signals against later payments before using them to select cases.

Check how the fields were built before using them as rules

The source is unchanged. The report uses all rows; no transaction-ID duplicates were found.

Check	Count	What it means
Negative supplied intervals	2,051,331	Recomputed from timestamps after sorting each sender.
Missing supplied intervals	896,513	All occur in rows labelled legitimate; check how this field was built.
Same-device consecutive sender payments	0	No unchanged-device comparison is available in sender histories.
Distinct fraud-type labels	1	All fraud labels say card_not_present; this is an outcome field.
January 2024 records	6,417	Only 1 January is present; excluded from the full-month comparison.
Duplicate transaction IDs	0	No rows were removed as duplicate transaction IDs.

For review planning

Use payment size to understand value under review. Investigate the city-change association with a complete account history. Confirm label and feature construction before treating any pattern as a prevention rule.

The supplied time interval reproduces differences in CSV row order, measured in hours; negative values are not chronological waiting times. The report recalculates sender gaps from timestamps. Before operational use, verify dataset provenance and feature definitions, then test proposed rules on later-dated data and count the false alerts.